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PROJECT ON

GROCERY STORE MANGEMENT

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GROCERY STORE MANGEMENT

AGENDA

- Introduction
- Objective of the Project
- Problem statement
- Tables
- ER Diagram and schema explanation
- SQL query results with screenshots
- Insights
- Recommendations
- Summary

INTRODUCTION:

- **Grocery Store Management System (GSM)** is a database-driven application designed to handle the daily operations of a grocery business.
- It allows easy management of **customers, suppliers, employees, products, categories, and orders**.
- Using **SQL as the backbone**, the system ensures that all information is stored in a structured and secure way.
- GSM helps in **tracking inventory levels**, managing **supplier details**, and monitoring **customer purchases**.
- It reduces errors from manual record-keeping and increases **accuracy, efficiency, and reliability**.
- Store managers can quickly generate reports like **sales history, most purchased products, and supplier performance**.
- Overall, GSM improves **business decision-making, customer satisfaction, and operational efficiency**.

OBJECTIVES:

- To design a **centralized SQL database** for storing and managing all grocery store information.
- To **simplify customer, supplier, employee, and product management**.
- To enable **easy tracking of orders, sales, and transactions**.
- To provide **accurate inventory management** and avoid stock-out/overstock issues.
- To improve **data accuracy, reliability, and security** compared to manual record-keeping.
- To generate useful **reports and insights** for better business decision-making.
- To save time, reduce manual errors, and ensure **efficient store operations**.

PROBLEM STATEMENT

- Grocery stores generate large amounts of data every day from products, suppliers, customer purchases, and orders. However, without a structured system to organize and analyze this information, valuable insights often remain hidden. This leads to challenges in identifying best-selling products, tracking loyal customers, managing inventory efficiently, and understanding revenue trends.
- The problem, therefore, is the lack of an integrated database and analytical framework that can transform raw transactional data into actionable business intelligence. Such a system should not only store and manage data effectively but also enable complex SQL-driven analysis—such as joins, aggregations, and filtering—to uncover patterns in sales, customer behavior, supplier performance, and profitability.

TABLE'S

1.Suppliers Table

Field	Type	Null	Key	Default	Extra
sup_id	tinyint	NO	PRI	NULL	
sup_name	varchar(255)	YES		NULL	
addess	text	YES		NULL	

2. Categories Table

Field	Type	Null	Key	Default	Extra
cat_id	tinyint	NO	PRI	NULL	
cat_name	varchar(255)	YES		NULL	

- 3. Employees Table

Field	Type	Null	Key	Default	Extra
emp_id	tinyint	NO	PRI	NULL	
emp_name	varchar(255)	YES		NULL	
hire_date	varchar(255)	YES		NULL	

4. Customers Table

Field	Type	Null	Key	Default	Extra
cust_id	smallint	NO	PRI	NULL	
cust_name	varchar(255)	YES		NULL	
address	text	YES		NULL	

5. Product Table

Field	Type	Null	Key	Default	Extra
prod_id	tinyint	NO	PRI	NULL	
prod_name	varchar(255)	YES		NULL	
sup_id	tinyint	YES	MUL	NULL	
cat_id	tinyint	YES	MUL	NULL	
price	decimal(10,2)	YES		NULL	

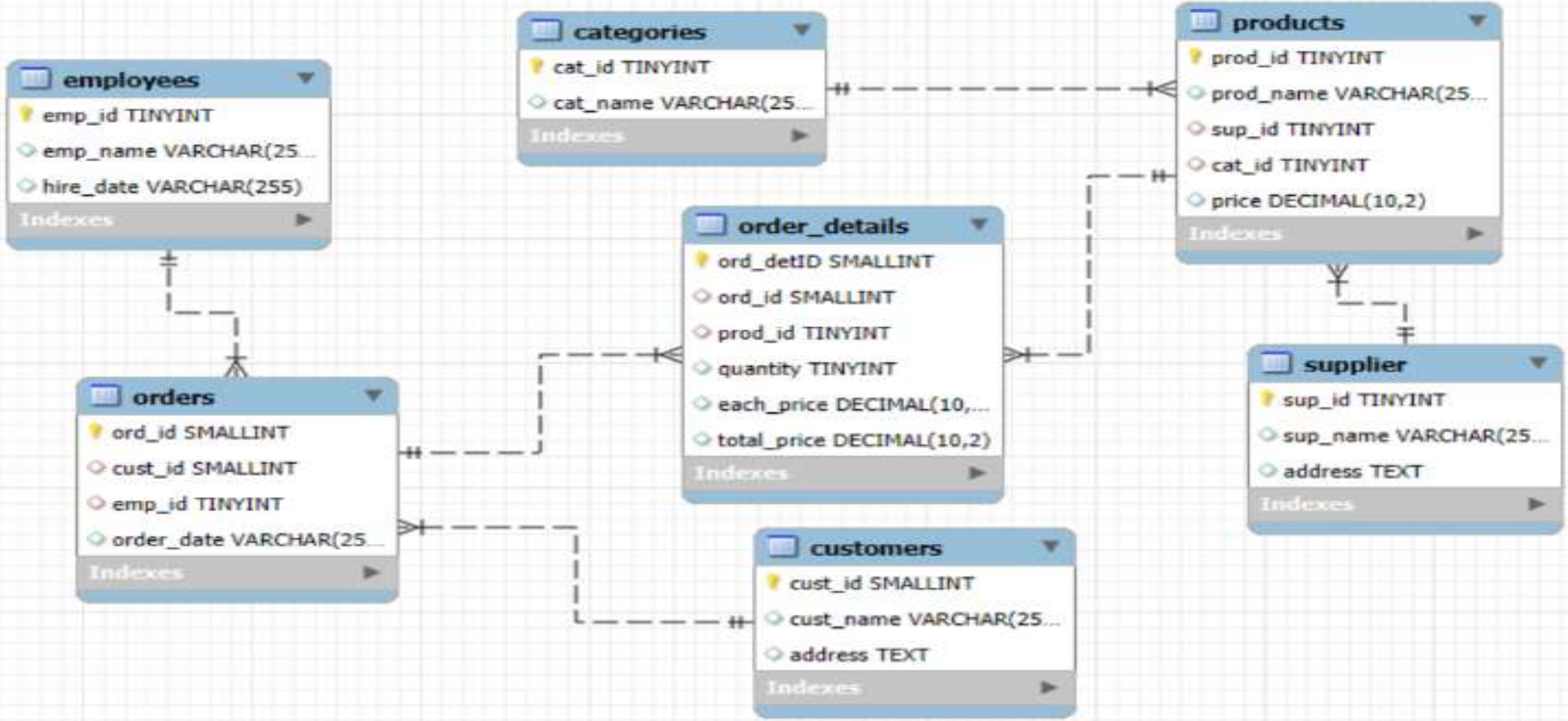
6. Orders Table

Field	Type	Null	Key	Default	Extra
ord_id	smallint	NO	PRI	NULL	
cust_id	smallint	YES	MUL	NULL	
emp_id	tinyint	YES	MUL	NULL	
order_date	varchar(255)	YES		NULL	

- 7. Order-details

Field	Type	Null	Key	Default	Extra
ord_detID	smallint	NO	PRI	NULL	auto_increment
ord_id	smallint	YES	MUL	NULL	
prod_id	tinyint	YES	MUL	NULL	
quantity	tinyint	YES		NULL	
each_price	decimal(10,2)	YES		NULL	
total_price	decimal(10,2)	YES		NULL	

ER DIAGRAM:


categories
cat_id TINYINT
cat_name VARCHAR(25...
Indexes
products
prod_id TINYINT
prod_name VARCHAR(25...
sup_id TINYINT
cat_id TINYINT
price DECIMAL(10,2)
Indexes
employees
emp_id TINYINT
emp_name VARCHAR(25...
hire_date VARCHAR(255)
Indexes
order_details
ord_detID SMALLINT
ord_id SMALLINT
prod_id TINYINT
quantity TINYINT
each_price DECIMAL(10,...
total_price DECIMAL(10,2)
Indexes
supplier
sup_id TINYINT
sup_name VARCHAR(25...
address TEXT
Indexes
orders
ord_id SMALLINT
cust_id SMALLINT
emp_id TINYINT
order_date VARCHAR(25...
Indexes
customers
cust_id SMALLINT
cust_name VARCHAR(25...
address TEXT
Indexes

1. CUSTOMER INSIGHTS

1. How many unique customers have placed orders

```
SELECT COUNT(DISTINCT cust_id) AS unique_customers
FROM orders;
```

	unique_customers
▶	156

2.Which customers have placed the highest number of orders?

```sql
select c.cust_id,c.cust_name,count(o.ord_id) as orders_count
from customers c
join orders o on c.cust_id = o.cust_id
group by c.cust_id,c.cust_name
order by orders_count desc;
```

cust_id	cust_name	orders_count
165	Jyotika	7
61	Aditi Rao	6
19	Chetan Naidu	5
32	Eshwar Menon	5
128	Hari Naidu	5
145	Chetan Rao	5
195	Amit Saxena	5
3	Chetan Rao	4

3. What is the total and average purchase value per customer?

```sql
SELECT
    c.cust_id,
    c.cust_name,
    SUM(od.total_price) AS total_purchase_value,
    AVG(od.total_price) AS average_purchase_value
FROM customers c
JOIN orders o
    ON c.cust_id = o.cust_id
JOIN order_details od
    ON o.ord_id = od.ord_id
GROUP BY c.cust_id, c.cust_name
ORDER BY total_purchase_value DESC;
```

cust_id	cust_name	total_purchase_value	average_purchase_value
19	Chetan Naidu	11256.82	1125.682000
166	Kapila	11099.51	1109.951000
67	Eshwar Rao	10819.96	901.663333
61	Aditi Rao	10230.64	1023.064000
7	Eshwar Iyer	9188.45	656.317857

4. Who are the top 5 customers by total purchase amount?

```
select c.cust_id,c.cust_name,sum(od.total_price) as total_spent
from customers c
join orders o on c.cust_id = o.cust_id
join order_details od on o.ord_id = od.ord_id
group by c.cust_id,c.cust_name
order by total_spent desc
limit 5;
```

cust_id	cust_name	total_spent
19	Chetan Naidu	11256.82
166	Kapila	11099.51
67	Eshwar Rao	10819.96
61	Aditi Rao	10230.64
7	Eshwar Iyer	9188.45

2.PRODUCT PERFORMANCE

1. How many products exist in each category?

```
select cat.cat_name,count(p.prod_id) as product_count
from categories cat
left join products p on cat.cat_id = p.cat_id
group by cat.cat_name;
```

cat_name	product_count
Grains & Cereals	18
Dairy Products	6
Snacks & Confectioneries	17
Personal Care	6
Household	3

2. What is the average price of products by category

```
select cat.cat_name,avg(p.price) as avg_price
from categories cat
join products p on cat.cat_id = p.cat_id
group by cat.cat_name;
```

	cat_name	avg_price
▶	Grains & Cereals	287.673333
	Dairy Products	366.943333
	Snacks & Confectioneries	278.892353
	Personal Care	364.991667
	Household	363.336667

3.Which products have the highest total sales volumes (by quantity)

```
select p.prod_name,sum(od.quantity) as total_qty
from products p
join order_details od on p.prod_id = od.prod_id
group by p.prod_name
order by total_qty desc;
```

prod_name	total_qty
Bath Soap	60
Hand Sanitizer	56
Dishwashing Soap	54
Potato Chips	54
Biscuits	54

Result 13

4. What is the total revenue genarated by each product?

```
select p.prod_name,sum(od.total_price) as revune
from products p
join order_details od on p.prod_id = od.prod_id
group by p.prod_name
order by revune desc;
```

prod_name	revune
Hand Sanitizer	27787.76
Biscuits	20995.92
Moong Dal	19695.02
Toothpaste	19688.95
Mustard Seeds	19516.68

Result 14

5.How do product sales vary by category and suppliers.

```
select c.cat_name,s.sup_name,
sum(od.quantity) as qty_sold,
sum(od.total_price) as total_revune
from products p
join categories c on p.cat_id = c.cat_id
join supplier s on p.sup_id = s.sup_id
join order_details od on p.prod_id = od.prod_id
group by c.cat_name, s.sup_name
order by total_revune desc;
```

cat_name	sup_name	qty_sold	total_revune
Personal Care	Aarya	205	69378.41
Grains & Cereals	Aarya	226	67701.10
Snacks & Confectioneries	Aarya	196	65538.71
Snacks & Confectioneries	Suresh	272	65307.14
Dairy Products	Sai	121	50740.60

Result 15

3. SALES AND ORDER TRENDS

1. How many orders have been placed in total?

```
SELECT COUNT(*) AS TotalOrders
FROM Orders;
```

	TotalOrders
▶	300

2. What is the average value per order?

```sql
SELECT AVG(OrderTotal) AS AvgOrderValue
FROM (
    SELECT ord_id, SUM(total_price) AS OrderTotal
    FROM order_details
    GROUP BY ord_id
) AS OrderValues;
```

	AvgOrderValue
▶	2153.632539

3. On which dates were the most orders placed?

```sql
SELECT order_date, COUNT(*) AS TotalOrders
FROM orders
GROUP BY order_date
ORDER BY TotalOrders DESC;
```

order_date	TotalOrders
9/10/2022	4
3/30/2022	4
1/30/2022	3
4/22/2022	3
1/14/2022	3

Result 19

4. What are the montly trends I order volume and revenue

```sql
SELECT left(o.order_date,7) as order_Month,
       COUNT(DISTINCT o.ord_id) AS order_volume,
       SUM(od.quantity * od.each_price) AS TotalRevenue
FROM orders o
JOIN order_details od ON o.ord_id = od.ord_id
GROUP BY order_Month
ORDER BY order_Month;
```

order_Month	order_volume	TotalRevenue
1/1/202	1	711.73
1/10/20	1	133.51
1/11/20	2	2217.14
1/12/20	1	3489.72
1/14/20	2	3333.23

Result 20

5. How do order patterns vary across weekdays and weekends?

```sql
SELECT
    CASE
        WHEN DAYOFWEEK(STR_TO_DATE(order_date, '%Y-%m-%d')) IN (1,7) THEN 'Weekend'
        ELSE 'Weekday'
    END AS DayType,
    COUNT(*) AS TotalOrders
FROM orders
GROUP BY DayType;
```

	DayType	TotalOrders
▶	Weekday	300

4. SUPPLIER CONTRIBUTION

1. How many suppliers are there in the database?

```
SELECT COUNT(*) AS TotalSuppliers
FROM supplier;
```

	TotalSuppliers
▶	5

2.Which supplier provides the most products?

```sql
SELECT s.sup_id, s.sup_name, COUNT(p.prod_id) AS ProductCount
FROM supplier s
JOIN products p ON s.sup_id = p.sup_id
GROUP BY s.sup_id, s.sup_name
ORDER BY ProductCount DESC
LIMIT 1;
```

	sup_id	sup_name	ProductCount
▶	3	Aarya	18

3.What is the average price of products from each suppliers

```
SELECT s.sup_id, s.sup_name, AVG(p.price) AS AvgProductPrice
FROM supplier s
JOIN products p ON s.sup_id = p.sup_id
GROUP BY s.sup_id, s.sup_name
ORDER BY AvgProductPrice DESC;
```

sup_id	sup_name	AvgProductPrice
2	Sai	342.672000
3	Aarya	319.326667
5	Karthik	288.225556
4	Suresh	281.818000
1	Aarav Sharma	271.366667

Result 24

4. Which suppliers contribute the most to total product sales (by revenue)

```sql
SELECT s.sup_id, s.sup_name,
       SUM(od.total_price) AS TotalRevenue
FROM supplier s
JOIN products p ON s.sup_id = p.sup_id
JOIN order_details od ON p.prod_id = od.prod_id
GROUP BY s.sup_id, s.sup_name
ORDER BY TotalRevenue DESC;
```

sup_id	sup_name	TotalRevenue
3	Aarya	221137.83
2	Sai	113588.51
4	Suresh	101688.78
5	Karthik	81861.96
1	Aarav Sharma	33052.85

Result 25

5. EMPLOYEE PERFORMANCE

1. How many employees have processed orders?

```
SELECT COUNT(DISTINCT emp_id) AS ActiveEmployees
FROM orders;
```

	ActiveEmployees
▶	10

2. Which employees have handled the most orders?

```sql
SELECT e.emp_id, e.emp_name,
       COUNT(DISTINCT o.ord_id) AS OrdersHandled
FROM employees e
JOIN orders o ON e.emp_id = o.emp_id
GROUP BY e.emp_id, e.emp_name
ORDER BY OrdersHandled DESC;
```

emp_id	emp_name	OrdersHandled
8	Diya Sharma 1	38
2	Aditya Singh 1	37
9	Arjun Kumar 1	32
3	Pari Kumar 1	31
5	Pari Sharma 1	31

Result 27

3. What is the total sales values processed by each employee?

```sql
SELECT e.emp_id, e.emp_name,
       SUM(od.total_price) AS TotalSales
FROM employees e
JOIN orders o ON e.emp_id = o.emp_id
JOIN order_details od ON o.ord_id = od.ord_id
GROUP BY e.emp_id, e.emp_name
ORDER BY TotalSales DESC;
```

emp_id	emp_name	TotalSales
2	Aditya Singh 1	79252.29
6	Zara Verma 1	71562.76
8	Diya Sharma 1	67241.85
3	Pari Kumar 1	66818.39
9	Arjun Kumar 1	54018.31

Result 28

4.What is the average order value handled per employee?

```sql
SELECT e.emp_id, e.emp_name,
       AVG(OrderValue) AS AvgOrderValue
FROM employees e
JOIN orders o ON e.emp_id = o.emp_id
JOIN (
    SELECT ord_id, SUM(total_price) AS OrderValue
    FROM order_details
    GROUP BY ord_id
) AS OrderVals ON o.ord_id = OrderVals.ord_id
GROUP BY e.emp_id, e.emp_name
ORDER BY AvgOrderValue DESC;
```

emp_id	emp_name	AvgOrderValue
1	Aarav Kumar 1	2768.572632
6	Zara Verma 1	2650.472593
2	Aditya Singh 1	2330.949706
3	Pari Kumar 1	2227.279667
7	Vihaan Singh 1	2112.081739

Result 29

6. ORDER DETAILS DEEP DRIVE

1. Relationship between quantity ordered and total price

```
SELECT quantity, total_price
FROM order_details;
```

quantity	total_price
3	421.87
1	441.95
4	665.06
2	438.73
4	1544.71

2. Average quantity ordered per product

```sql
SELECT p.prod_id, p.prod_name, AVG(od.quantity) AS AvgQuantity
FROM products p
JOIN order_details od ON p.prod_id = od.prod_id
GROUP BY p.prod_id, p.prod_name
ORDER BY AvgQuantity DESC;
```

	prod_id	prod_name	AvgQuantity
▶	40	Butter	4.5556
	31	Toothpaste	3.6667
	46	Potato Chips	3.6000
	42	Tomato Ketchup	3.5000
	22	Mustard Seeds	3.4615

3. How does the unit price vary across products and orders?

```
SELECT p.prod_id, p.prod_name, od.each_price
FROM products p
JOIN order_details od ON p.prod_id = od.prod_id
ORDER BY p.prod_name, od.each_price;
```

prod_id	prod_name	each_price
11	Almonds	315.57
11	Almonds	315.57
11	Almonds	315.57
11	Almonds	315.57
11	Almonds	315.57

INSIGHTS

- Customer Insights
- Product Performance
- Sales & Revenue Trends
- Employee Performance
- Suppliers Contribution
- Order Details

RECOMANDATION

Enhance Customer Loyality:

Launch loyalty programs for top customers . Offer personalized discounts based on buying history.

Boost Product Sales:

Increase stock and promotion for best-selling items . Introduce bundle offers for slow-moving products.

Optimize Inventory:

Maintain safety stock for high-demand products . Reduce overstock of low-performing items.

Improve Supplier Relationships:

Prioritize reliable suppliers with consistent delivery . Negotiate better terms with high-performing suppliers.

Data-Driven Decision Making:

Continue using SQL reports for weekly/monthly insights . Integrate sales trends into marketing and stocking strategies.

SUMMARY

- The Grocery Store Management System organizes and analyzes store data using SQL, enabling quick and accurate reporting.
- It provides actionable insights to boost sales, manage inventory efficiently, and improve overall business decisions.

ANY QUESTIONS ?

Thank You!